Chapter 1. The Genesis of Fourier Analysis

The vibrating string and the flow of heat

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1.1 The vibrating string

Motivation. Fourier analysis was born from two physical problems: the vibrating string and the flow of heat. Both lead to linear partial differential equations whose separated solutions are sines and cosines. The general solution is then a superposition, i.e. a Fourier series, and the central question becomes: which functions can be written this way?

Simple harmonic motion. A mass m on a spring with constant k obeys $-ky(t) = my''(t)$. With $c = \sqrt{k/m}$ the solutions are $y(t) = a\cos ct + b\sin ct = A\cos(ct-\varphi)$, with **amplitude** $A = \sqrt{a^2+b^2}$, **period** $2\pi/c$ and **frequency** $c/2\pi$; here $a = y(0)$, $b = y'(0)/c$. In complex form $e^{ix} = \cos x + i\sin x$ (Euler), $\cos x = \frac{e^{ix}+e^{-ix}}{2}$, $\sin x = \frac{e^{ix}-e^{-ix}}{2i}$.

Standing and travelling waves. A **travelling wave** has the form $u(x,t) = F(x-ct)$ (a profile moving with speed c). A **standing wave** has the form $u(x,t) = \varphi(x)\psi(t)$: the profile keeps its shape and only its amplitude varies in time.

1.1.1 Derivation of the wave equation

Model a string of length L, density ρ and tension τ fixed at its ends by N particles of mass ρh ($h = L/N$) at $x_n = nh$, connected by springs. Let $y_n(t)$ be the vertical displacement of the n-th particle. Hooke's law for the vertical components of the tensions gives

$$\rho h\, y_n''(t) = \frac{\tau}{h}\big(y_{n+1}(t) + y_{n-1}(t) - 2y_n(t)\big).$$

As $h \to 0$ and $y_n(t) \to u(x,t)$, the right side divided by h tends to $\tau\,\partial_x^2 u$, giving the **wave equation**

$$\frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, \qquad c = \sqrt{\tau/\rho}. \tag{1.1}$$

(Changing $x \mapsto x/L$, $t \mapsto ct/L$ we may take $c = 1$, $L = \pi$.) **Invariance properties:** the equation is linear and invariant under translations in x and t.

1.1.2 Solution to the wave equation

d'Alembert's solution. Introduce $\xi = x+t$, $\eta = x-t$ and $v(\xi,\eta) = u(x,t)$. The wave equation becomes $\frac{\partial^2 v}{\partial\xi\partial\eta} = 0$, so $v = F(\xi) + G(\eta)$:

$$u(x,t) = F(x+t) + G(x-t),$$

the sum of two travelling waves. On the real line with $u(x,0) = f(x)$, $\partial_t u(x,0) = g(x)$:

$$u(x,t) = \frac{f(x+t)+f(x-t)}{2} + \frac{1}{2}\int_{x-t}^{x+t} g(y)\,dy. \tag{1.2}$$

Check [Supplement]: $F+G = f$ and $F'-G' = g$ give $F = \frac{f}{2} + \frac{1}{2}\int_0^x g + C$, $G = \frac{f}{2} - \frac{1}{2}\int_0^x g - C$. On $[0,\pi]$ with $u(0,t) = u(\pi,t) = 0$, extend f, g to $[-\pi,\pi]$ as odd functions and then 2π-periodically; (1.2) then satisfies the boundary conditions.

Superposition of standing waves. Separating variables, $u(x,t) = \varphi(x)\psi(t)$ gives $\frac{\psi''}{\psi} = \frac{\varphi''}{\varphi} = \lambda$, a constant. Boundedness forces $\lambda = -m^2 \le 0$, so $\psi(t) = A\cos mt + B\sin mt$ and $\varphi(x) = \tilde{A}\cos mx + \tilde{B}\sin mx$; $\varphi(0) = \varphi(\pi) = 0$ forces $\varphi(x) = \sin mx$ with $m = 1, 2, \dots$. The standing waves are

$$u_m(x,t) = (A_m\cos mt + B_m\sin mt)\sin mx,$$

the m-th **harmonic** ($m=1$: **fundamental tone**; $m\geq 2$: **overtones**, of higher frequency $m/2\pi$). By linearity

$$u(x,t)=\sum_{m=1}^{\infty}(A_m\cos mt+B_m\sin mt)\sin mx \tag{1.3}$$

is a candidate solution, and the initial condition $u(x,0)=f(x)$ requires

$$f(x)=\sum_{m=1}^{\infty}A_m\sin mx. \tag{1.4}$$

Fourier's question: can every (reasonable) f on $[0,\pi]$ with $f(0)=f(\pi)=0$ be written as in (1.4)? If so, the orthogonality $\int_0^\pi \sin mx\sin nx\,dx=\frac{\pi}{2}\delta_{mn}$ forces

$$A_m=\frac{2}{\pi}\int_0^\pi f(x)\sin mx\,dx,$$

and similarly B_m is determined by $\partial_t u(x,0)=g(x)=\sum mB_m\sin mx$. Extending f oddly to $[-\pi,\pi]$, (1.4) is the special case of

$$F(x)=\sum_{m=-\infty}^{\infty}a_m e^{imx},\qquad a_m=\frac{1}{2\pi}\int_{-\pi}^{\pi}F(x)e^{-imx}\,dx. \tag{1.5}$$

1.1.3 Example: the plucked string

Pluck the string at $p\in(0,\pi)$ to height h with zero initial velocity: $f(x)=\frac{xh}{p}$ for $0\leq x\leq p$, $f(x)=\frac{h(\pi-x)}{\pi-p}$ for $p\leq x\leq\pi$, $g=0$. Then $B_m=0$ and, integrating by parts,

$$A_m=\frac{2h\sin mp}{m^2p(\pi-p)},\qquad u(x,t)=\sum_{m\geq 1}A_m\cos mt\sin mx.$$

Since $\sum|A_m|<\infty$, the series converges uniformly, and u is continuous. In general, though, (1.3) may fail to be twice differentiable, so it solves the wave equation only in a generalised sense (compare d'Alembert's formula).

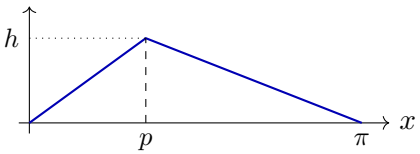


Figure. *The plucked string at $t=0$.*

1.2 The heat equation

Motivation. Heat flow leads to the heat and Laplace equations. Steady-state temperatures in a disc are again obtained by separating variables, this time in polar coordinates, and lead to Fourier series in the angle θ.

1.2.1 Derivation of the heat equation

For a plate $\mathbb{R}^2$ with temperature $u(x,y,t)$, specific heat σ and thermal conductivity κ: the heat in a small square S is $\sigma\iint_S u$, and the heat flux across ∂S is $\kappa\int_{\partial S}\partial u/\partial n$ (Fourier's law: heat flows from

hotter to colder, proportionally to the gradient). Equating the rate of change of heat in S with the flux, dividing by the area and shrinking S:

$$\frac{\partial u}{\partial t} = \frac{\kappa}{\sigma}\Big(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\Big) = \frac{\kappa}{\sigma}\,\triangle u. \tag{1.6}$$

In a **steady state** $\partial u/\partial t = 0$, and u satisfies the **Laplace equation** $\triangle u = 0$; its solutions are **harmonic functions**.

1.2.2 Steady-state heat equation in the disc

Dirichlet problem. Find u with $\triangle u = 0$ in the unit disc $D = \{x^2 + y^2 < 1\}$ and $u = f$ on the circle C. In polar coordinates

$$\triangle u = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2}.$$

Separating $u = F(r)G(\theta)$, $\frac{r^2F'' + rF'}{F} = -\frac{G''}{G} = \lambda$. G is 2π-periodic, so $\lambda = m^2$ and $G(\theta) = e^{\pm im\theta}$, $m \geq 0$. Then $r^2F'' + rF' - m^2F = 0$ has solutions $F = r^{\pm m}$ $(m \geq 1)$, $F = 1, \log r$ $(m = 0)$; boundedness at 0 leaves $F = r^{|m|}$. So the candidate solution is

$$u(r,\theta) = \sum_{m=-\infty}^{\infty} a_m r^{|m|} e^{im\theta}, \qquad f(\theta) = \sum a_m e^{im\theta}. \tag{1.7}$$

Substituting $a_m = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(\varphi)e^{-im\varphi}d\varphi$ and summing the geometric series formally,

$$u(r,\theta) = \frac{1}{2\pi}\int_{-\pi}^{\pi} P_r(\theta - \varphi) f(\varphi)\, d\varphi, \qquad P_r(\gamma) = \sum_{m\in\mathbb{Z}} r^{|m|}e^{im\gamma} = \frac{1 - r^2}{1 - 2r\cos\gamma + r^2},$$

the **Poisson kernel**; this is made rigorous in Chapter 2.

1.3 Selected exercises

Motivation. Five exercises from the book that justify the steps taken for granted above: the form of the separated solutions, the discrete-to-continuous limit in the wave equation, the polar Laplacian, and the plucked-string coefficients.

Exercise 1.6 (Solutions of $f'' + c^2 f = 0$)

Let $c > 0$ and let f be a C^2 function on $\mathbb{R}$ with $f''(t) + c^2 f(t) = 0$. Prove that there are constants a, b with $f(t) = a\cos ct + b\sin ct$ for all t.

Solution.

Step 1. Let $g(t) = f(t)\cos ct - c^{-1}f'(t)\sin ct$ and $h(t) = f(t)\sin ct + c^{-1}f'(t)\cos ct$, both C^1. Using $f'' = -c^2 f$,

$$g'(t) = f'\cos ct - cf\sin ct - c^{-1}f''\sin ct - f'\cos ct = -cf\sin ct + cf\sin ct = 0,$$
$$h'(t) = f'\sin ct + cf\cos ct + c^{-1}f''\cos ct - f'\sin ct = cf\cos ct - cf\cos ct = 0.$$

Step 2. By the mean value theorem $g \equiv a$ and $h \equiv b$ are constant on $\mathbb{R}$ (with $a = f(0)$, $b = f'(0)/c$).

Step 3. Then, for every t,

$$a\cos ct + b\sin ct = f(\cos^2 ct + \sin^2 ct) + c^{-1}f'(-\sin ct\cos ct + \sin ct\cos ct) = f(t).$$

This justifies the form of ψ and φ in the separation of variables of §1.1.2. ∎

Exercise 1.8 (Second difference quotient)

Let F be C^2 on (a,b) and $x \in (a,b)$. Show that for $x+h \in (a,b)$

$$F(x+h) = F(x) + hF'(x) + \tfrac{h^2}{2}F''(x) + h^2\varphi(h), \qquad \varphi(h) \to 0 \ (h \to 0),$$

and deduce that $\big(F(x+h) + F(x-h) - 2F(x)\big)/h^2 \to F''(x)$.

Solution.

Step 1. By the fundamental theorem of calculus, $F(x+h) - F(x) = \int_x^{x+h} F'(y)\,dy$ and $F'(y) = F'(x) + \int_x^y F''(s)\,ds$. Writing $F''(s) = F''(x) + \big(F''(s) - F''(x)\big)$ and using $\int_x^{x+h}(y-x)\,dy = h^2/2$,

$$F(x+h) = F(x) + hF'(x) + \tfrac{h^2}{2}F''(x) + R(h), \qquad R(h) = \int_x^{x+h}\!\!\int_x^y \big(F''(s) - F''(x)\big)\,ds\,dy.$$

Step 2. Let $\omega(\delta) = \sup_{|s-x|\le\delta} |F''(s) - F''(x)|$; $\omega(\delta) \to 0$ as $\delta \to 0$ by continuity of F''. For y between x and $x+h$ the inner integral is at most $|y-x|\,\omega(|h|)$, so $|R(h)| \le \omega(|h|)\,\big|\int_x^{x+h} |y-x|\,dy\big| = \tfrac{h^2}{2}\omega(|h|)$ (for either sign of h).

Step 3. Set $\varphi(h) = R(h)/h^2$ for $h \neq 0$. Then $|\varphi(h)| \le \tfrac{1}{2}\omega(|h|) \to 0$.

Step 4. Adding the expansions at h and $-h$, the first-order terms cancel:

$$\frac{F(x+h) + F(x-h) - 2F(x)}{h^2} = F''(x) + \varphi(h) + \varphi(-h) \to F''(x).$$

This is the limit used to pass from the particle model to the wave equation (1.1). ∎

Exercise 1.9 (Missing harmonics of the plucked string)

For the plucked string of §1.1.3, derive $A_m = \frac{2h \sin mp}{m^2 p(\pi - p)}$. For which p are the 2nd, 4th, 6th, … harmonics missing? For which p are the 3rd, 6th, 9th, … harmonics missing?

Solution.

Step 1. Since f is continuous, piecewise linear and $f(0) = f(\pi) = 0$, integrating by parts (the boundary terms vanish) gives

$$\int_0^\pi f(x)\sin mx\,dx = \frac{1}{m}\int_0^\pi f'(x)\cos mx\,dx = \frac{1}{m}\Big(\frac{h}{p}\int_0^p \cos mx\,dx - \frac{h}{\pi - p}\int_p^\pi \cos mx\,dx\Big).$$

Step 2. The two integrals are $\frac{\sin mp}{m}$ and $-\frac{\sin mp}{m}$, so the bracket equals $\frac{h\sin mp}{m}\big(\frac{1}{p} + \frac{1}{\pi - p}\big) = \frac{\pi h \sin mp}{m\,p(\pi - p)}$. Multiplying by $\frac{2}{\pi}$ gives the stated A_m.

Step 3. The m-th harmonic is missing iff $A_m = 0$ iff $\sin mp = 0$. All even harmonics vanish iff $\sin 2kp = 0$ for all $k \ge 1$; taking $k = 1$ this forces $2p \in \pi\mathbb{Z}$, i.e. $p = \pi/2$ for $p \in (0,\pi)$, and conversely $p = \pi/2$ gives $\sin(k\pi) = 0$. (The odd harmonics remain, since $\sin(m\pi/2) = \pm 1$ for odd m.)

Step 4. Similarly the multiples of 3 vanish iff $\sin 3p = 0$, i.e. $p = \pi/3$ or $p = 2\pi/3$. Plucking at the midpoint or at a third of the string removes exactly those overtones. ∎

Exercise 1.10 (Laplacian in polar coordinates)

Let u be C^2 on an open set of $\mathbb{R}^2 \setminus \{0\}$ and $v(r,\theta) = u(r\cos\theta, r\sin\theta)$. Prove

$$u_{xx} + u_{yy} = v_{rr} + \frac{1}{r}v_r + \frac{1}{r^2}v_{\theta\theta}, \qquad |u_x|^2 + |u_y|^2 = |v_r|^2 + \frac{1}{r^2}|v_\theta|^2.$$

Solution. Write $c = \cos\theta$, $s = \sin\theta$; all derivatives of u are evaluated at (rc, rs).

Step 1 (first derivatives). By the chain rule $v_r = cu_x + su_y$ and $v_\theta = -rsu_x + rcu_y$.

Step 2 (gradient identity). Hence $\frac{1}{r}v_\theta = -su_x + cu_y$ and

$$|v_r|^2 + \tfrac{1}{r^2}|v_\theta|^2 = |cu_x + su_y|^2 + |-su_x + cu_y|^2 = (c^2+s^2)(|u_x|^2 + |u_y|^2),$$

since the cross terms $2cs\,\mathrm{Re}(u_x\bar{u}_y)$ and $-2cs\,\mathrm{Re}(u_x\bar{u}_y)$ cancel.

Step 3 (second derivatives). Differentiating Step 1 again and using $u_{xy} = u_{yx}$ ($u \in C^2$),

$$\begin{aligned} v_{rr} &= c^2u_{xx} + 2csu_{xy} + s^2u_{yy}, \\ v_{\theta\theta} &= -rcu_x - rsu_y + r^2(s^2u_{xx} - 2csu_{xy} + c^2u_{yy}) = -rv_r + r^2(s^2u_{xx} - 2csu_{xy} + c^2u_{yy}). \end{aligned}$$

Step 4. Therefore $v_{rr} + \frac{1}{r}v_r + \frac{1}{r^2}v_{\theta\theta} = (c^2+s^2)u_{xx} + (s^2+c^2)u_{yy} = u_{xx} + u_{yy}$; the mixed terms and the v_r terms cancel. ∎

Exercise 1.11 (The Euler equation $r^2F'' + rF' - n^2F = 0$)

Let $n \in \mathbb{Z}$. Show that the twice differentiable solutions on $r > 0$ of $r^2F''(r) + rF'(r) - n^2F(r) = 0$ are exactly the linear combinations of r^n, r^{-n} if $n \neq 0$, and of $1, \log r$ if $n = 0$.

Solution. The equation depends only on n^2, so assume $n \geq 0$.

Step 1 (these are solutions). For $F = r^{\pm n}$: $r^2(\pm n)(\pm n - 1)r^{\pm n-2} + r(\pm n)r^{\pm n-1} - n^2r^{\pm n} = (n^2 \mp n \pm n - n^2)r^{\pm n} = 0$. For $F = \log r$: $r^2(-r^{-2}) + r \cdot r^{-1} = 0$; constants are clearly solutions. By linearity so are their combinations.

Step 2 (reduction). Let F be a solution and $g(r) = r^{-n}F(r)$, twice differentiable. From $F = r^ng$,

$$r^2F'' + rF' - n^2F = r^n\left[r^2g'' + 2nrg' + n(n-1)g + rg' + ng - n^2g\right] = r^{n+1}\left[rg'' + (2n+1)g'\right],$$

so $rg'' + (2n+1)g' = 0$ on $(0, \infty)$.

Step 3. Then $\left(r^{2n+1}g'\right)' = r^{2n}(rg'' + (2n+1)g') = 0$, so by the mean value theorem on the interval $(0, \infty)$, $r^{2n+1}g'(r) = C$ for a constant C.

Step 4. If $n \geq 1$: $g' = Cr^{-2n-1}$ gives $g = -\frac{C}{2n}r^{-2n} + D$ and $F = Dr^n - \frac{C}{2n}r^{-n}$. If $n = 0$: $g' = C/r$ gives $F = g = C\log r + D$. This completes the classification used in §1.2.2 (and in the uniqueness part of Theorem 5.7). ∎